Prototype Implementation of a Low-Cost Force Measurement System for Neuro-Rehab Applications

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Introduction

Stroke is one of the leading causes of death and disability in the Philippines. According to the Philippine Statistics Authority (2023), cerebrovascular diseases, including stroke, accounted for 11.9% of total deaths, making it the second leading cause of mortality in the country. Survivors often face long-term impairments such as reduced grip strength, poor hand coordination, and difficulty performing daily activities. Rehabilitation is essential, but access to advanced rehabilitation devices is limited due to high costs and the scarcity of locally developed solutions.

Existing rehabilitation technologies, such as robotic exo-gloves and grip force monitoring systems, are available in developed countries but remain expensive and inaccessible to most Filipino patients. Imported devices can cost hundreds of thousands of pesos, placing them far beyond the reach of public hospitals and rural health centers. This creates a pressing need for affordable, locally developed rehabilitation technologies that can provide effective therapy and monitoring.

This study proposes the prototype implementation of a low-cost force measurement system using an nRF52840 and a Force Sensitive Resistor (FSR). The system focuses on grip force monitoring for neuro-rehabilitation applications, specifically for stroke patients. By detecting whether the applied force is too weak or too strong, the system can provide feedback during rehabilitation exercises. Furthermore, this technology can serve as a foundation for future applications such as bionic arms and prosthetic devices, where force sensing is critical for safe and functional use.

Objectives

To design and implement a low-cost pressure sensing system using Nordic nRF52840 and Force Sensitive Resistor (FSR) sensors.

To calibrate the sensor with known weights and establish a measurable relationship between applied force and sensor output.

To demonstrate real-time data collection and visualization of grip force using VSCode and Excel.

To validate the system's ability to detect excessive or insufficient force during simulated rehabilitation tasks, ensuring safe application in neuro-rehabilitation.

To highlight the potential of this system as a foundation for scalable biomedical projects, including robotic exo-gloves, rehabilitation devices, and most importantly, future bionic arm applications.

To establish groundwork for later integration of Artificial Intelligence (AI), which can enhance prosthetic and rehabilitation systems by enabling adaptive thresholds, personalized feedback, and intelligent control mechanisms.

Methodology

Materials & Equipment

Microcontroller: nRF52840
Sensors: 2x Circular FSR sensors
Resistors: 4x 5.1kΩ resistors (2 per FSR, in parallel configuration)
Software: PlatformIO in VS Code for nRF52840 programming
Data Output: Python script to save serial data to CSV files

Procedure

1. PlatformIO Setup for nRF52840

A basic Arduino sketch is uploaded via PlatformIO that reads both FSR analog pins (A0 and A1). The code implements a 6-sample averaging routine for noise reduction and outputs (timestamp, fsr1_value, fsr2_value) in CSV format via Serial. Sampling rate is set to (10Hz) (100ms delay between samples) with automatic timestamping using the `millis()` function. An LED heartbeat indicator blinks at 2Hz to confirm system operation.

2. Python CSV Data Collection

A Python script uses pyserial to read the CSV-formatted data stream from the nRF52840. The script automatically creates timestamped CSV files, parsing the incoming time_ms, fsr1, fsr2 data. The raw ADC values (0-1023) from both FSRs are logged with their original millisecond timestamps directly from the microcontroller.

3. Execution

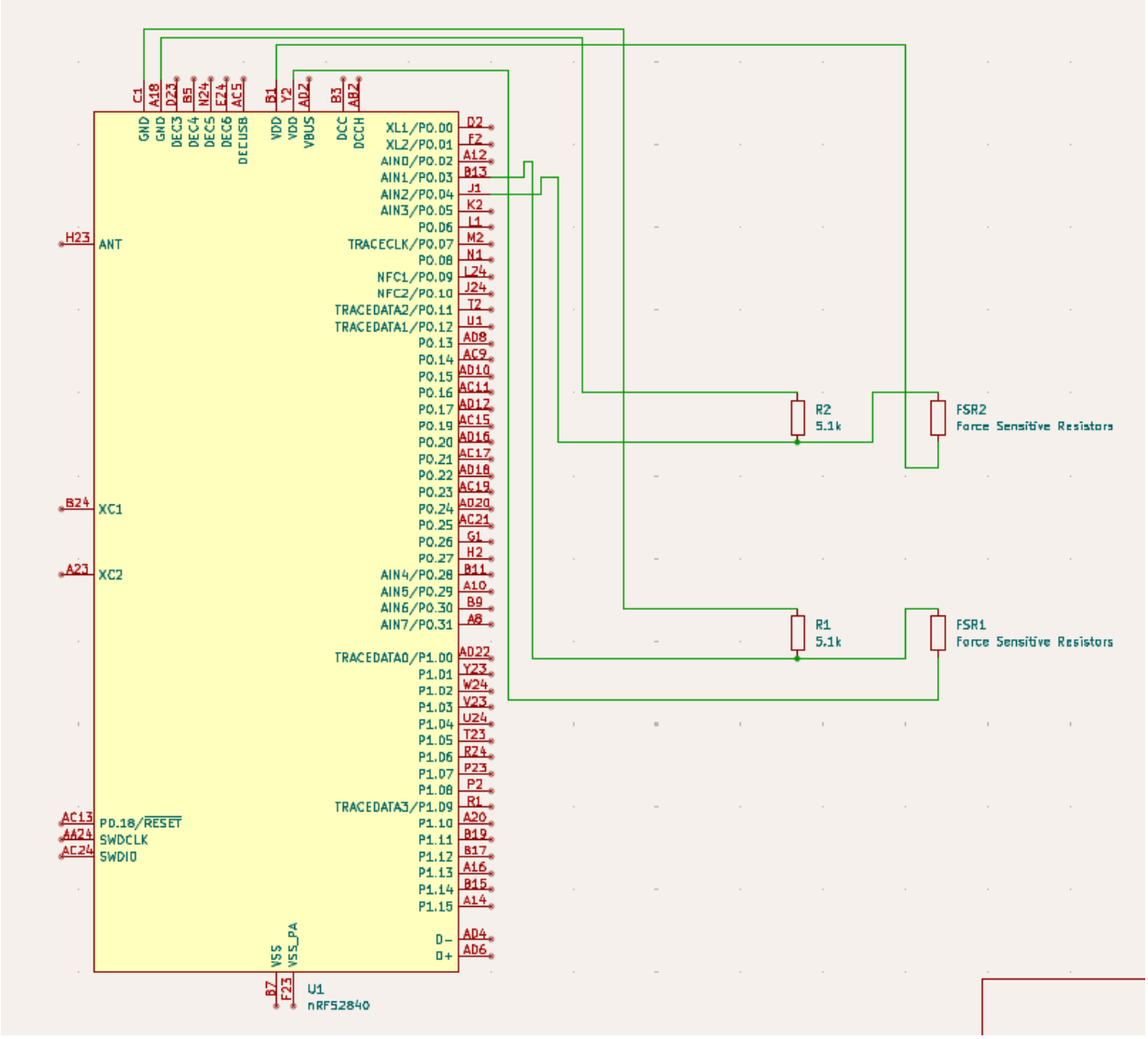
Upload the PlatformIO code to nRF52840
Run the Python scriptApply pressure to FSR sensors to generate dataCSV file is automatically saved when stopping the script

Current Status

The nRF52840 with dual FSR sensors is functionalCSV data collection is working

System provides raw data ready for future stroke rehabilitation analysis

Circuit Diagrams



Program/Code

platform.ini

```
[env:nrf52840_dk]
platform = nordicnrf52
board = nrf52840_dk
framework = arduino
# Use 115200 for faster, reliable continuous monitoring
monitor_speed = 115200
# Remove a hard-coded upload port so PlatformIO auto-detects the correct interface.
# upload_port = COM9  ; CHANGE TO YOUR ACTUAL PORT
```

main.cpp

```
#include <Arduino.h>


const uint8_t FSR_PIN1 = A0;
const uint8_t FSR_PIN2 = A1;


const unsigned long SAMPLE_MS = 100;
const int SAMPLES = 6;


const unsigned long HEARTBEAT_MS = 500;

void setup() {

  Serial.begin(115200);


  unsigned long start = millis();
  while (!Serial && (millis() - start) < 2000) {
    delay(5);
  }


  pinMode(LED_BUILTIN, OUTPUT);


  Serial.println("time_ms,fsr1,fsr2");
}

int readAverage(uint8_t pin) {
  long sum = 0;
  for (int i = 0; i < SAMPLES; ++i) {
    sum += analogRead(pin);

    delay(1);
```

```
  }
  return (int)(sum / SAMPLES);
}

void loop() {
  static unsigned long lastSample = 0;
  static unsigned long lastHeartbeat = 0;

  unsigned long now = millis();


  if (now - lastHeartbeat >= HEARTBEAT_MS) {
    digitalWrite(LED_BUILTIN, (now / HEARTBEAT_MS) & 1);
    lastHeartbeat = now;
  }

  if (now - lastSample >= SAMPLE_MS) {
    lastSample = now;

    int v1 = readAverage(FSR_PIN1);
    int v2 = readAverage(FSR_PIN2);


    Serial.print(now);
    Serial.print(',');
    Serial.print(v1);
    Serial.print(',');
    Serial.println(v2);
  }
}
```

Significance of the Study

This project addresses a critical healthcare need in the Philippines, where stroke remains the second leading cause of death, accounting for 11.9% of total deaths in 2022 (Philippine Statistics Authority, 2023). Survivors often require long-term rehabilitation to regain grip strength and hand coordination. However, most Filipino patients cannot access advanced rehabilitation devices due to their high cost and limited availability. Imported robotic gloves and prosthetic systems are prohibitively expensive, restricting their use to a few specialized hospitals and research institutions.

By developing a low-cost, locally implementable force measurement system, this study provides:

- Affordable rehabilitation tools that can be deployed in hospitals, clinics, and even home-based therapy.
- A foundation for more advanced systems such as robotic exo-gloves and prosthetic bionic arms, where grip force monitoring is essential.
- A step toward Philippine-made biomedical devices, reducing dependence on expensive imports and fostering local expertise in biomedical engineering.

Although some rehabilitation technologies exist in the country, their presence is limited and insufficient to meet the growing demand. The majority of patients, especially in rural areas, have little to no access to such devices. This highlights the urgent need for affordable, scalable solutions like the proposed system.

Furthermore, this project is not limited to stroke rehabilitation. The same force sensing framework can be applied to a wide range of biomedical engineering projects, most notably the development of bionic arms. In future iterations, AI can be integrated into such prosthetic systems to improve adaptability and patient-specific performance. Research has shown that AI and machine learning can significantly enhance prosthetic and orthotic rehabilitation by enabling intelligent control, adaptive feedback, and improved user interaction (Nayak & Das, 2020). This demonstrates that the proposed prototype is not only relevant for immediate rehabilitation needs but also serves as a strategic stepping stone toward advanced prosthetic technologies.

Conclusion

The prototype implementation of a low-cost force measurement system demonstrates that affordable grip force monitoring is feasible using readily available components such as nRF528340 or even an Arduino and FSR sensors. The system successfully measures applied force, detects excessive or insufficient pressure, and provides real-time feedback.

This project represents an important step toward accessible neuro-rehabilitation technologies in the Philippines, addressing the needs of stroke patients who require affordable and effective therapy tools. Furthermore, the same sensing principles can be expanded into exo-glove rehabilitation devices and bionic arm prosthetics, ensuring continuity from prototype to advanced biomedical applications.

References

Philippine Statistics Authority. (2023). *Causes of Death in the Philippines (Preliminary): 2022*.

Department of Health. *Stroke in the Philippines: Burden and Interventions*.

Interlink Electronics. *FSR Sensor Application Notes*.

Philippine Statistics Authority. (2023). *Causes of Death in the Philippines (Preliminary): 2022*.

Nayak, S., & Das, R. K. (2020). *Application of Artificial Intelligence (AI) in Prosthetic and Orthotic Rehabilitation*. IntechOpen. https://doi.org/10.5772/intechopen.93903

Lapresa, M., Lauretti, C., Scotto di Luzio, F., Bressi, F., Santacaterina, F., Bravi, M., Guglielmelli, E., Zollo, L., & Cordella, F. (2023). Development and validation of a system for the assessment and recovery of grip force control. Bioengineering, 10(1), 63.